Estimating Angles in Degrees

Year 5 · Mathematics · Measurement | B1 · Learn + Notice

Name: _____ Date: _____

LEARNING INTENTION

I can estimate angles using familiar degree benchmarks.

KEY IDEA

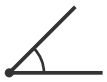
Estimate before measuring. Use 45°, 90°, 180°, 270° and 360° as benchmarks; estimates do not need to be exact.

VOCABULARY

estimate · degree · benchmark · acute · reflex

LEARN IT: A Worked Model

A small opening about halfway to a right angle is approximately 45°. An opening just beyond a right angle might be about 110°.



45°

Use the marked opening. Diagrams for calculation tasks are not drawn to scale.

SUCCESS CRITERIA

- ☐ I can estimate angles using familiar degree benchmarks.
- ☐ I can explain the benchmark, tool or angle rule I used.
- ☐ I can check my answer and correct a mistake.

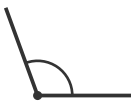
NOTICE IT: Guided Practice

Estimate each marked angle in degrees.

A.

Answer: _____

B.



Answer: _____

C.

Answer: _____

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PRACTISE IT: Independent Questions

Estimate each marked angle in degrees.

If needed, extend an arm with a ruler. Keep the vertex and opening unchanged.

A.

Answer: _____

B.

Answer: _____

C.

Answer: _____

D.

Answer: _____

E.

Answer: _____

F.

Answer: _____

APPLY IT

Choose two practice angles. Measure them with a protractor and compare your estimates. For a reflex angle, measure the smaller opening and subtract from 360°.

EXPLAIN HOW YOU KNOW: Misconception Check

A student estimates a clearly obtuse angle as 40°. Explain how a benchmark catches the mistake.

CHALLENGE: Reasoning Task

Sketch an angle you estimate as 135°. Explain how 90° and 180° helped.

MY SELF-REVIEW

Circle one: I understand it / I need more practice / I need help

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ANSWER KEY: Guided + Independent Practice

Notice targets: 45°, 110°, 270°. Practice targets: A 35°; B 85°; C 125°; D 175°; E 230°; F 300°. Accept reasonable estimates, typically within 15°; discuss wider estimates rather than treating them as exact measurement errors.

Apply: measured values within 2° of targets.

Reason: an obtuse angle must exceed 90°.

Challenge: approximately midway between 90° and 180°.

COMMON MISCONCEPTION

An estimate is a reasoned approximation, not a measured value. Ask for the benchmark used.

TEACHER PROMPT

Ask: "Which opening are you using? What benchmark or angle relationship applies? How can you check your answer?"

NEXT-STEP GUIDANCE

Move to measuring acute and obtuse angles.

PRINTING AND ASSESSMENT

Print A4 at actual size (100%). Use a ruler and protractor. Accept ±2° for careful measurement and construction. Discuss estimates separately from measurement accuracy.

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